Task 1

CJD

2025-11-28

Librarys:

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
```

Set WD and import data

```
# CHANGE THIS TO YOUR OWN WD
setwd("/Users/charliedicker/Documents/Uni/Year 1/R basics (data science)/Coursework/Exported_data")

gdp_per_capita_df <- read.csv("gdp_per_capita_df.csv")
```

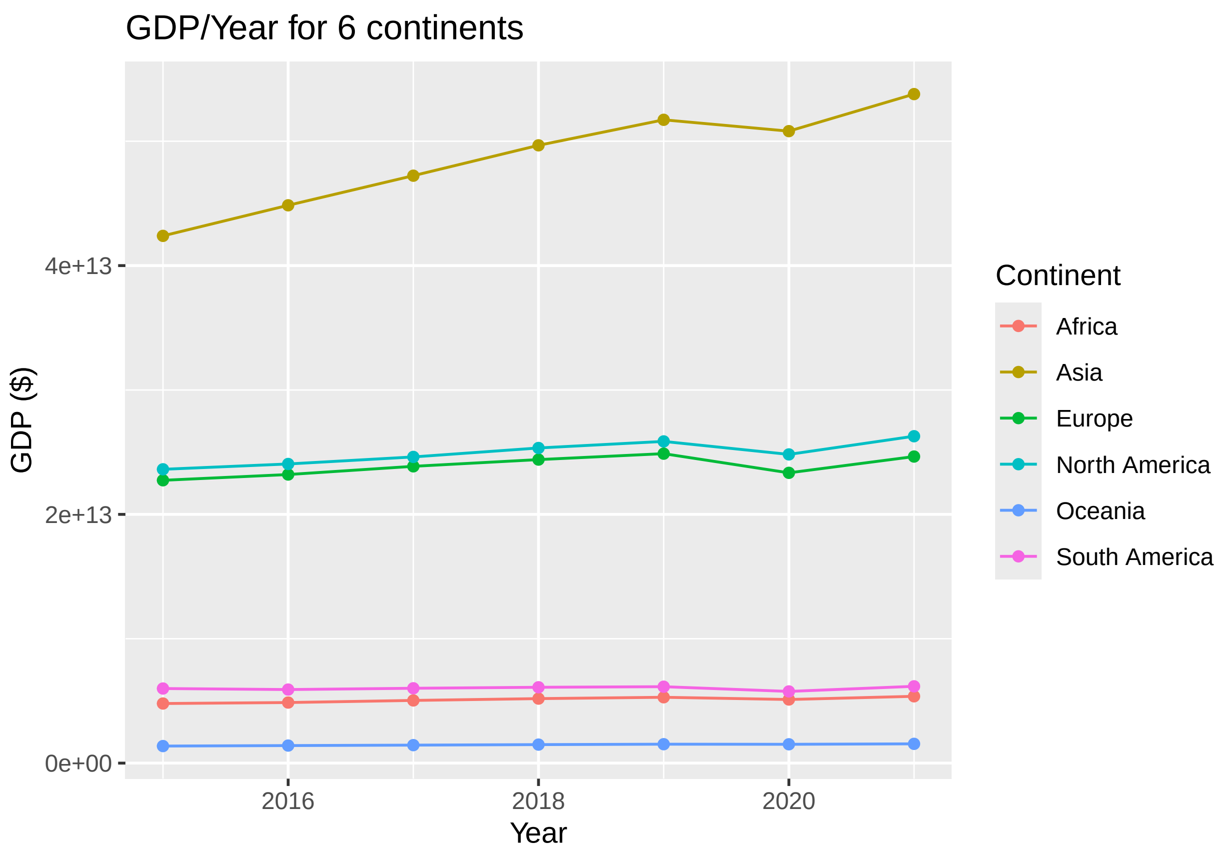
Calculate the total GDP of each continent

```
continent_gdp_totals <- gdp_per_capita_df %>%
  group_by(Continent, Year) %>%
  summarise(
    Total_GDP = sum(Gdp, na.rm = TRUE)
  ) %>%
  ungroup()
```

```
## `summarise()` has grouped output by 'Continent'. You can override using the
## `.groups` argument.
```

Plot a graph of GDP for each continent, vs time

```
continent_gdp_totals %>% ggplot(aes(x = Year, y = Total_GDP, colour = Continent)) +
  geom_point() +
  geom_line() +
  labs(title = "GDP/Year for 6 continents", y = "GDP ($)")
```

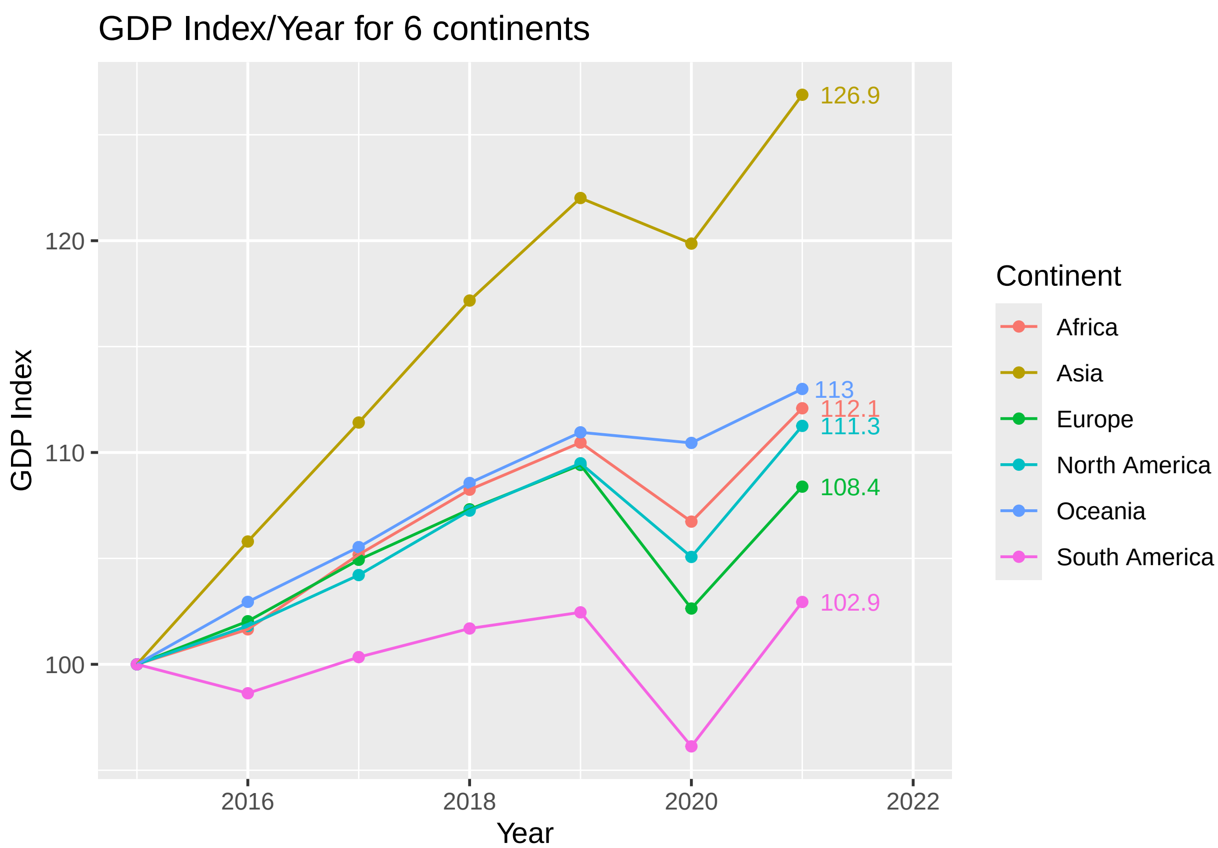


I decided this graph could show more if they all started at an index of 100

```
continent_gdp_totals <- continent_gdp_totals %>%
  group_by(Continent) %>%
  mutate(
    GDP_Index = Total_GDP / first(Total_GDP) * 100
  )
```

We now replot the graph, using normalised values

```
continent_gdp_totals %>% ggplot(aes(x = Year, y = GDP_Index, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_text(
  data = continent_gdp_totals %>% filter(Year == 2021),
  aes(label = round(GDP_Index, 1)),
  hjust = -0.3,
  size = 3,
  show.legend = FALSE) +
  expand_limits(x = 2022) +
  labs(title = "GDP Index/Year for 6 continents", y = "GDP Index")
```



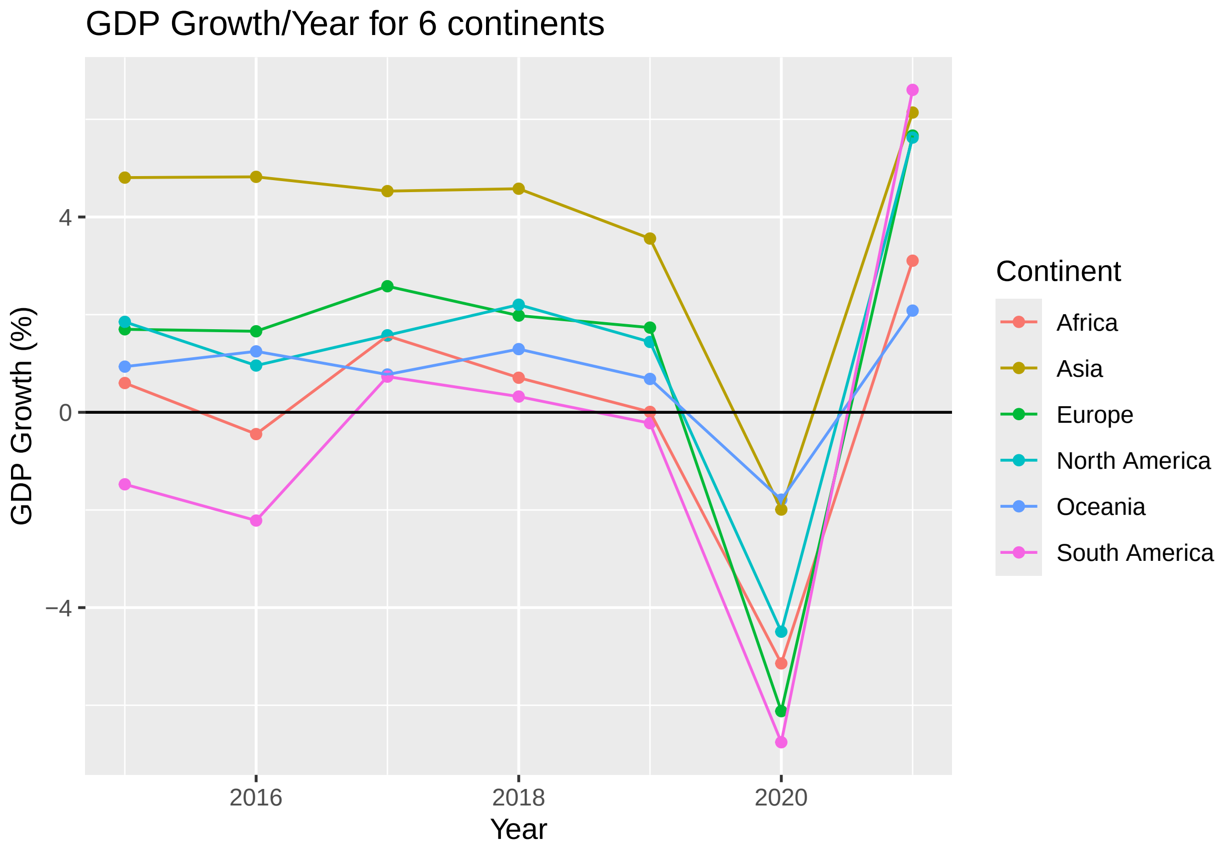
Calculate the weighted growth of each continent

```
continents_growth_df <- gdp_per_capita_df %>%
  group_by(Continent, Year) %>%
  summarise(
    Weighted_Growth = sum(Growth * Gdp, na.rm = TRUE) / sum(Gdp, na.rm = TRUE)
  ) %>%
  ungroup()
```

```
## `summarise()` has grouped output by 'Continent'. You can override using the
## `.groups` argument.
```

Plot the growth graphs of each continent

```
continents_growth_df %>% ggplot(aes(x = Year, y = Weighted_Growth, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 0) +
  labs(title = "GDP Growth/Year for 6 continents", y = "GDP Growth (%)")
```



REF:

28/11/2025 - asked ChatGPT - “i have text near the last points on the x axis, how can i make it so it doesn’t overlap the axis”